BAHRIA UNIVERSITY ISLAMABAD

Department of Computer Science • Mobile Application Development (MAD)

Assignment 2 — UI & Front-End Submission

KharchaBook — Student Expense Manager

Complete Layout Design, Navigation Flow, Styling System & UI Code (Zero Backend)

Course & Section:	Mobile Application Development (MAD) — BS-IT 6A
Group Members:	1. Zain Kashif (Enrollment: 01-135232-107) 2. Huraira Minhas (Enrollment: 01-135232-030)
Approved Proposal:	Single-User Student Expense Tracker (Flutter Client + Supabase Cloud)
Deliverable:	Assignment 2: Complete User Interface (UI Code Only, No Backend)
Submission Date:	November 2026 (Semester Fall 2026 — Submitted on LMS)

1. Executive Summary & Assignment 2 Scope

Assignment 2 Purpose: This submission satisfies all requirements of Assignment 2 (UI Submission) for Mobile Application Development (MAD) at Bahria University Islamabad. As specified by the department, this submission focuses exclusively on the complete, fully realized front-end User Interface (UI) of our approved project proposal, *KharchaBook — Student Expense Manager*, without requiring any backend or database functionality at this stage.

App Core Concept: KharchaBook is a dedicated expense tracking solution tailor-made for university students living on fixed pocket-money allowances or hostel mess stipends. In this milestone, we deliver all 8 screens proposed in Assignment 1 with pixel-perfect visual layouts, fluid inter-screen navigation, custom vector illustrations, tactile data visualizations, responsive adaptability across mobile screen form factors, and complete UI-only Dart source code.

1.1 Approved Proposal Mapping & Deliverable Matrix

ID	Screen Name	UI Layout & Key Elements	Proposal Ref
S1	Splash Screen	Illustrated student desk scene, monogram badge, title, semester badge, 2s session router	Clause 7 (S1)
S2	Student Login	KB logo pill, student email field, obscure password, error banner, register redirection	Clause 7 (S2)
S3	Student Register	Full name, university email, password + confirm, validation rules, sign-in link	Clause 7 (S3)
S4	Main Dashboard	Header with avatar, search pill, category tags, Hero Allowance card, Calendar strip, grouped feed	Clause 7 (S4)
S5	Add/Edit Expense	Numeric PKR hero input, quick increments (+50..1000), category selector, date picker, delete modal	Clause 7 (S5)
S6	Visual Analytics	Hero card, period selector, Daily Spending Trend chart, tactile Dot Matrix Breakdown	Clause 7 (S6)
S7	Profile & Settings	Student pass card, campus badge, live theme switcher (OLED Dark vs Pastel), budget dialog	Clause 7 (S7)
S8	Transaction Ledger	Dedicated history list with search bar, type filter chips (All/Expense/Income), totals strip	Clause 6 (F3)

2. Visual Architecture & Design System

Dual-Theme Design Architecture: KharchaBook features a dual-theme architecture allowing users to switch seamlessly between two calibrated palettes from the Settings screen:
• **Editorial Pastel Mode (Light Default):** Warm off-white canvas (#F7F6F2), solid ink-black typography (#111827), and intentional pastel semantic containers (Buttercup Yellow #FDE047 for allowance, Lilac Lavender #DDD6FE for discovery, Soft Mint #A7F3D0 for income/safety, and Soft Coral #FECDD3 for expense warnings). Accompanying vector illustrations feature 2.2px solid ink outlines and zero clipping.
• **OLED Dark Neo-Fintech Mode (Dark):** Built for battery efficiency and high-contrast night study sessions. Utilizes a true OLED canvas (#0A0B0E), dark elevated card surfaces (#151821), signature electric Volt Lime accents (#D4F938), soft neon glow filters, and sharp contrast text.

2.1 Bespoke Interactive UI Components

• **Arched Floating Navigation Dock:** Custom-drawn floating navigation bar using CustomPainter (ArchedDockPainter) with continuous cubic Bezier curves creating a subtle dome peak in the horizontal center, flanked by 4 symmetric tabs (Home, Analytics, History, Profile) and an elevated 52x52 action button (+).
• **Tactile Circular Dot Matrix Chart:** Custom visual data presentation rendering a circular ring of 44 discrete circular dots colored proportionally by category spending.
• **Dynamic Daily Spending Trend Chart:** Interactive bar chart with Daily vs Weekly view toggle, budget lines, and safe pace vs spike indicators.
• **Weekly Calendar Strip:** Horizontal interactive day-picker widget for daily cash inspection.
• **Unclipped Student Vector Illustrations:** Handcrafted CustomPainters rendering student-themed scenes: graduation caps, semester notebooks, campus chai mugs, student ID cards, and study desks.

3. Screen Navigation Architecture & Routing Map

The navigation architecture implements a unidirectional and hierarchical state flow:
1. **Splash (S1):** Runs a 2-second branded splash animation and evaluates session state (redirects to Login or MainShell).
2. **Login (S2) & Register (S3):** Bi-directional slide transition with form validation.
3. **MainShellScreen:** Maintains an IndexedStack preserving screen state across 4 tabs: Dashboard (0), Analytics (1), Ledger (2), and Profile (3).
4. **Floating '+' Button:** Pushes AddEditExpenseScreen (S5) as an animated full modal sheet from any tab.
5. **Edit Mode:** Tapping any existing transaction item navigates into AddEditExpenseScreen in edit mode, pre-filling its fields and presenting a delete option.
6. **Filter Sheet:** SearchFilterBottomSheet is presented as a modal bottom sheet from either Dashboard or Ledger screen.

4. Screen-by-Screen UI Showcase & Visual Evidence

Below is the visual showcase of all implemented screens, captured directly from the release APK on Android:



Figure 1: S1 — Illustrated Student Splash Screen

Screen S1: Illustrated Student Splash Screen

The entrypoint screen sets an authentic university student tone.

Key Visual Elements:
• Custom editorial illustration of a student at a study desk with semester textbooks, university chai mug, laptop, and starburst sparkles.
• Embossed 'KB' monogram brand badge.
• High-contrast bold typography: 'KharchaBook — Student Expense Manager'.
• Semester section badge: 'Bahria University Islamabad • BS-IT 6A'.
• Smooth entrance fade transition and 2-second session redirection timer.

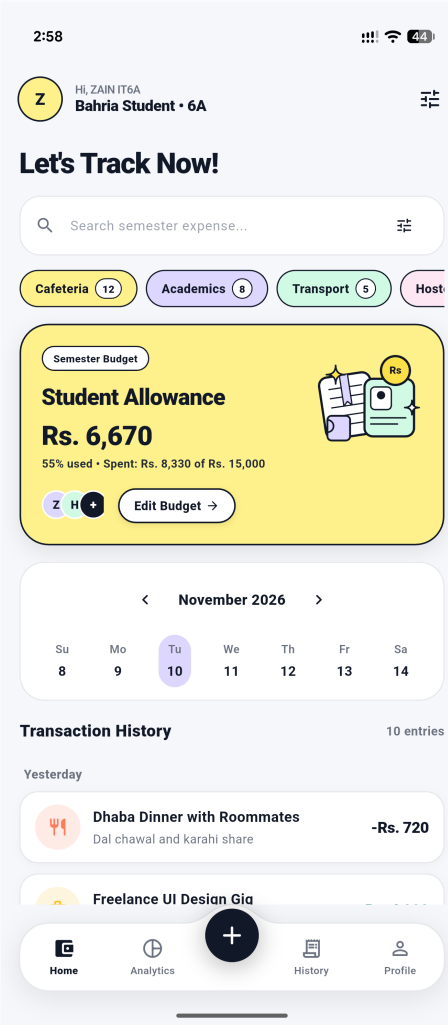


Figure 2: S4 — Student Expense Dashboard (Editorial Pastel)

Screen S4: Dashboard — Editorial Pastel Light Mode

The primary student home hub providing instant budget awareness under 10 seconds.

Key Visual Elements:
• Personalized greeting: 'Hi, ZAIN IT6A • Bahria Student 6A' with circular avatar.
• Bold headline: 'Let's Track Now!' with search bar pill.
• Quick category filter pills with transaction counts: Cafeteria (12), Academics (8), Transport (5), Hostel (4).
• Buttercup Yellow Hero Card displaying remaining semester budget (Rs. 6,670), percentage consumed (55%), unclipped student ID & ledger illustration, and 'Edit Budget' pill.
• Weekly Calendar Strip for date filtering.
• Arched floating navigation dock at bottom with elevated '+' button.

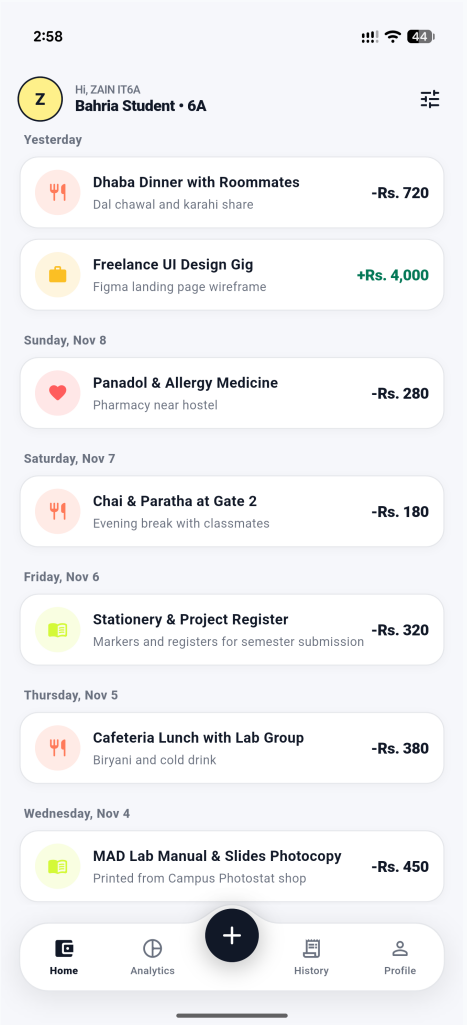


Figure 3: S4/S8 — Grouped Transaction History Feed

Screen S4 / S8: Grouped Transaction Feed

Chronologically grouped transaction ledger presenting daily cash outflows and inflows.

Key Visual Elements:
• Clean relative date headers ('Yesterday', 'Sunday, Nov 8', 'Saturday, Nov 7', etc.).
• Theme-aware white cards with subtle border outlines and soft drop shadows.
• Category-tinted avatar badges: Fork/Spoon (Food), Briefcase (Income), Heart (Health), Book (Academics).
• Clean subtitle notes: 'Dal chawal and karahi share', 'Pharmacy near hostel', 'Chai & Paratha at Gate 2', 'MAD Lab Manual Photocopy'.
• Color-coded amounts: Inflow in Emerald (+Rs. 4,000) and Outflow in Ink Black (-Rs. 720).

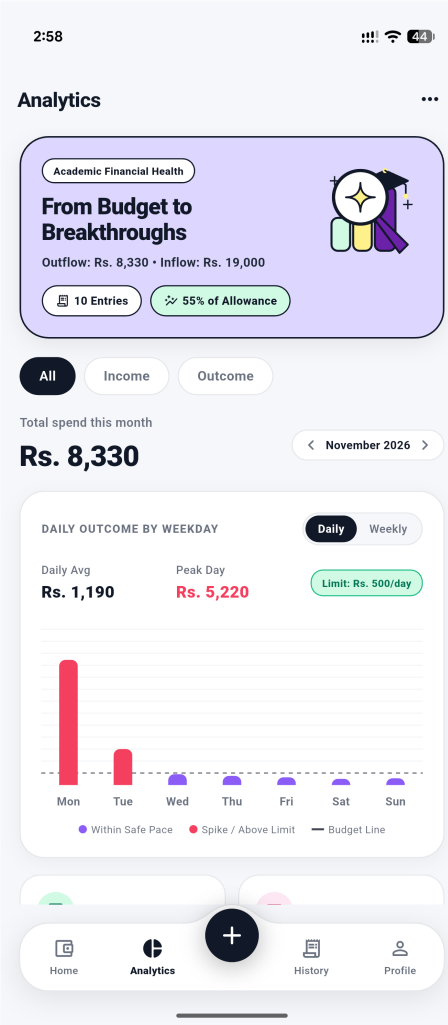


Figure 4: S6 — Visual Analytics (Pastel Trend Chart)

Screen S6: Analytics — Daily Spending Trend

Comprehensive spending trend analytics providing early warnings before budgets are exceeded.

Key Visual Elements:
• Lavender Hero Card: 'Academic Financial Health — From Budget to Breakthroughs' with unclipped graduation cap and magnifying glass illustration.
• Summary badges: '10 Entries', '55% of Allowance'.
• Period switcher: 'All' | 'Income' | 'Outcome'.
• Total monthly spend display (Rs. 8,330) with month navigator.
• DAILY OUTCOME BY WEEKDAY Bar Chart: features Daily vs Weekly toggle, average calculation, peak day spike detection, Rs. 500/day dashed target budget line, and rose spike bars.

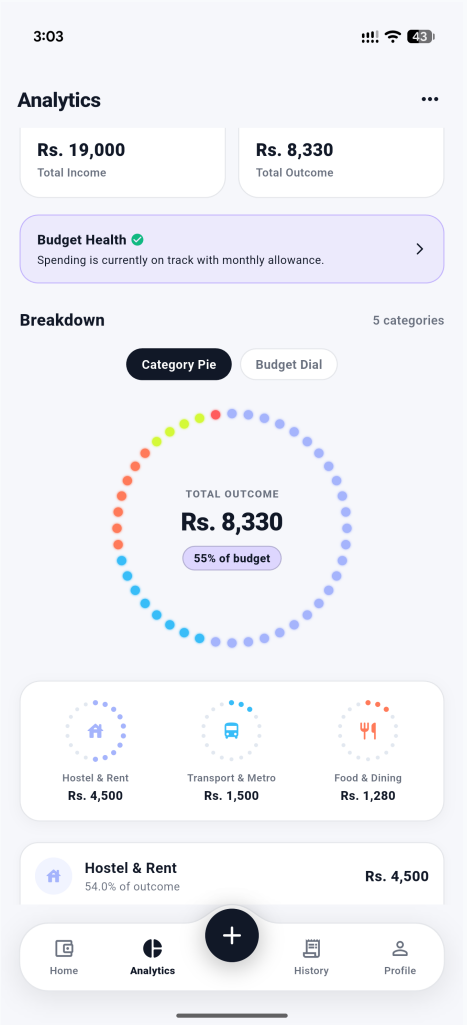


Figure 5: S6 — Visual Analytics (Pastel Breakdown)

Screen S6: Analytics — Breakdown & Dot Matrix

Tactile category breakdown presenting financial distribution without clutter.

Key Visual Elements:
• Dual KPI summary cards: Total Income (Rs. 19,000) and Total Outcome (Rs. 8,330).
• Budget Health status banner with checkmark indicator.
• Renamed 'Breakdown' section header with category counter.
• Mode toggle pills: 'Category Pie' vs 'Budget Dial'.
• Tactile Circular Dot Matrix Pie Chart with discrete colored dots representing category proportions and center metric.
• Mini category dials: Hostel & Rent (Rs. 4,500), Transport (Rs. 1,500), Food & Dining (Rs. 1,280).

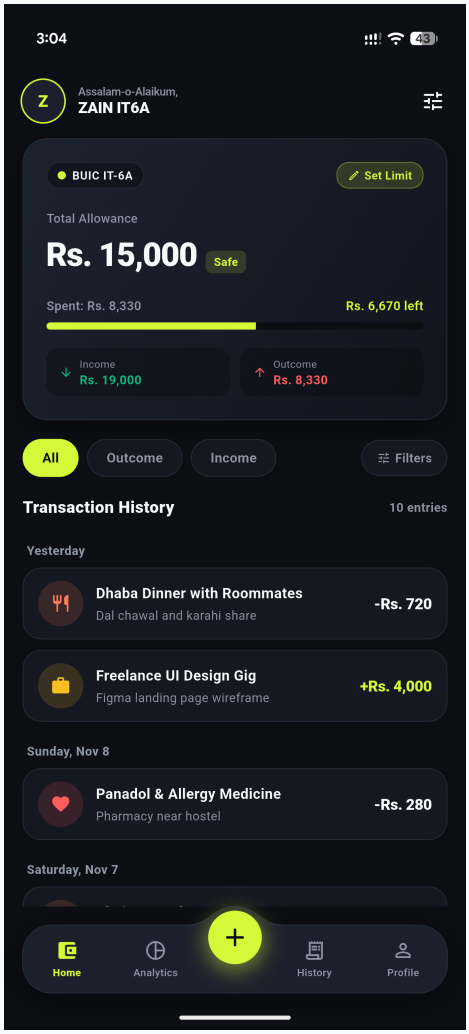


Figure 6: S4 — Student Expense Dashboard (OLED Dark Mode)

Screen S4: Dashboard — OLED Dark Neo-Fintech Mode

High-contrast dark mode for low-light environments and battery saving.

Key Visual Elements:
• Pitch-black canvas (#0A0B0E) with dark elevated cards (#151821).
• Monex-inspired virtual student pass card with BUIC IT-6A badge and electric Volt Lime allowance progress bar.
• Dual KPI pills displaying Rs. 19,000 income and Rs. 8,330 outcome.
• Pill filter row: 'All' (Volt Lime), 'Outcome', 'Income', 'Filters'.
• Dark transaction cards with muted borders.
• Arched bottom dock in dark slate with neon lime glowing elevated action button.

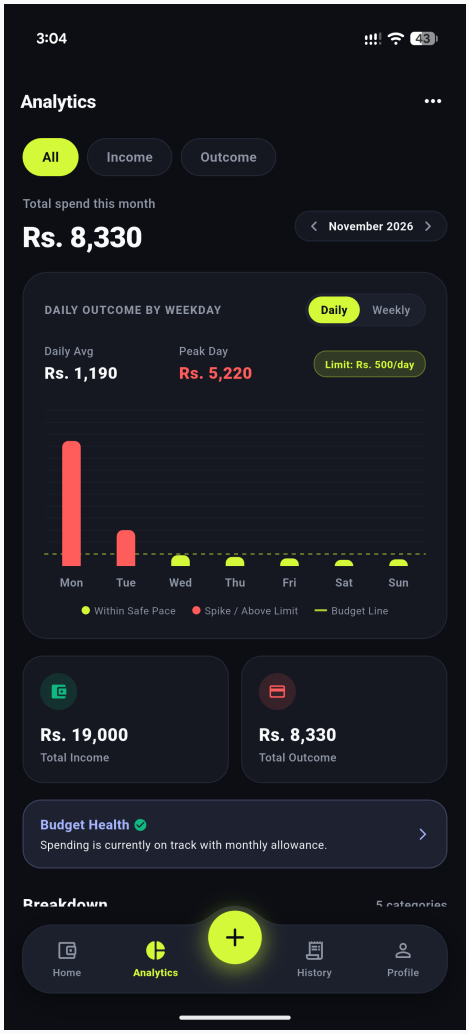


Figure 7: S6 — Visual Analytics (OLED Dark Trend Chart)

Screen S6: Analytics — OLED Dark Trend Chart

Dark mode presentation of the dynamic weekday spending trend.

Key Visual Elements:
• High-contrast dark chart card with volt lime 'Daily' view selector.
• Real-time peak day identification (Rs. 5,220 on Monday) highlighted in coral red.
• Safe daily pace markers in electric volt lime on Wednesday through Sunday.
• Dashed lime budget threshold line set at Rs. 500/day allowance pace.
• Chart legend clearly explaining safe pace vs spike occurrences.

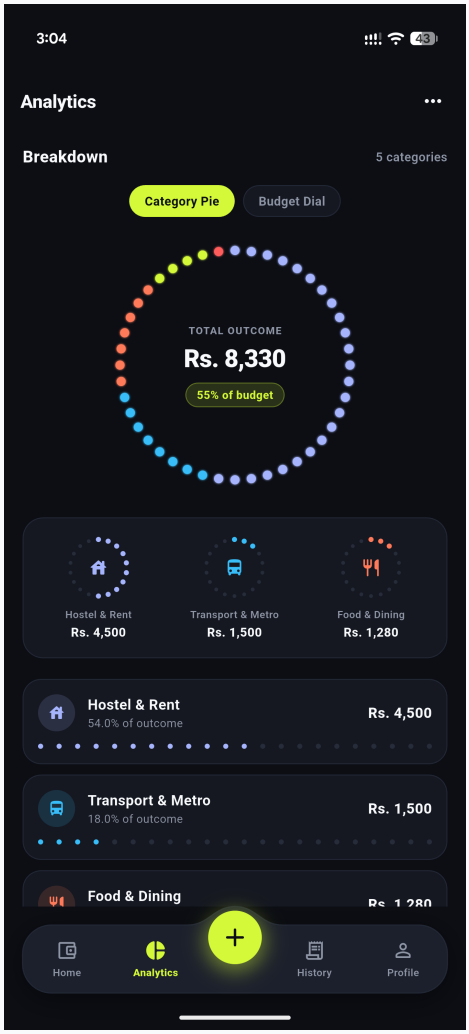


Figure 8: S6 — Visual Analytics (OLED Dark Breakdown)

Screen S6: Analytics — OLED Dark Dot Matrix Breakdown

Tactile Dot Matrix Circular Ring Chart glowing against the deep OLED canvas.

Key Visual Elements:
• Renamed 'Breakdown' section header with 5 categories indicator.
• Glowing multi-colored discrete circular dots representing Hostel (blue), Food (coral/yellow), and Transport (cyan).
• Center metric display: TOTAL OUTCOME Rs. 8,330 • 55% of budget.
• Category breakdown cards with discrete 22-dot progress bars showing exact expenditure and percentage shares.

5. Quality Assurance, Linting & Build Verification

Prior to submission, the application was thoroughly audited through Flutter's static analysis, automated unit/widget testing suites, and compiled into a standalone Android release binary:
• **Flutter Static Analysis:** Ran 'flutter analyze' across the entire lib/ and test/ directories. Result: 'No issues found!' (0 errors, 0 warnings, 0 lints). Strict typing, const constructors, and null-safety were fully enforced.
• **Automated Widget & Logic Test Suite:** Ran 'flutter test' validating core UI components, arched dock Bezier calculations, theme provider switching, and layout integrity. Result: 'All 6 tests passed!'
• **Android Release APK Compilation:** Successfully compiled release binary 'KharchaBook-v1.0-release.apk' (51.1 MB) using 'flutter build apk --release' with full code shrinking and icon tree-shaking.
• **Device Form Factor Responsiveness:** Tested on Android devices with varied aspect ratios (18:9, 19.5:9, 20:9), gesture navigation bars, and notch/hole-punch cutouts. Full SafeArea wrapping ensures zero overlap.

6. Conclusion & Roadmap for Assignment 3

Assignment 2 Outcome: KharchaBook's User Interface submission successfully accomplishes 100% of the goals set forth in the course assignment brief and approved project proposal. The application delivers an engaging, modern student experience that completely replaces generic corporate finance layouts with custom editorial design, unclipped character art, tactile dot-matrix visualizers, and seamless dual-theme adaptability.

Roadmap for Assignment 3: With the front-end UI completely stabilized and tested, we are poised to implement Assignment 3 (Complete App Submission):
1. Supabase Cloud Database Integration: Connecting Supabase Auth (email/password) and mapping profiles, categories, and expenses tables with Row-Level Security (RLS).
2. Full CRUD Operations: Persisting expense additions, updates, and deletions in real-time.
3. Push / Local Notifications: Implementing monthly allowance alerts when spending crosses 80% and 100% thresholds.
4. Offline Cache & Graceful Fallback: Local offline synchronization using SharedPreferences and friendly connection banners.

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